

Jacob Tenorio

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EDUCATION

Washington University in St. Louis

Degree Expected May 2027

B.S. in Mechanical Engineering | Germany Ballintyn Scholar | GPA: 3.51 | Dean's List: Fall 2023, Spring 2025, Fall 2025

Relevant Coursework: Fluid & Solid Mechanics, Heat Transfer, Thermal Systems Design, Machine Elements, Numerical Methods

EXPERIENCE

Rehab Robotics Lab, University of Illinois Chicago: *Research Assistant*

June – August 2025

- Shaped lab's roadmap for exoskeleton experiments by identifying physiological correlations through multivariate analysis
- Reduced identification time of desired frame ranges in motion-capture trials by 80% with MATLAB script
- Reduced manual file-renaming overhead by 85% with automated Python script
- Ensured timely approval of requested protocol modifications by drafting an institutional review board rebuttal summary

AbbVie | Business and Technology Solutions: *Intern*

August 2021 – August 2022

- Reduced supervisor's research time by independently investigating available technologies for call center system implementation

UL Solutions | Digital Security and Brand Protection: *Intern*

August 2019 – August 2021

- Contributed to company-wide cybersecurity awareness campaign by developing PowerPoints, posters, and social media content
- Extended cybersecurity campaign to audience of 300 students through an internet security presentation
- Reduced supervisor's research time by developing databases of grant opportunities for law enforcement training programs

PROJECTS

AIAA Design/Build/Fly & SAE Aero Design Competition Airplanes

August 2023 – Present

- Lead structures and testing teams designing competition-compliant aircraft via SolidWorks top-down modeling
- Design mission-specific components (e.g., removable wings, deployable glider, cargo & passenger compartments) in SolidWorks
- Validate aerodynamic & structural design choices through CFD and FEA simulations (e.g., rib spacing, motor mount, spar length)
- Reduced new-member CAD training time by 50% by designing an onboarding curriculum

Automotive HVAC Refrigerant Cycle Simulation & Lifecycle Impact Analysis

April – May 2026

- Modeled four automotive refrigerants across conventional and transcritical vapor-compression cycles in Python
- Optimized $CO_2/R-744$ transcritical high-side pressure to maximize COP, balancing cooling capacity against compressor work
- Conducted TEWI analysis, finding indirect emissions dominate climate impact, near-zero GWP insufficient to minimize footprint

Radial Heat Conduction in a Micro-Turbine Disk

April – May 2026

- Simulated steady-state radial conduction; COMSOL axisymmetric model, BCs: fixed rim T , forced convection, lubricant cooling
- Validated COMSOL against Bessel fin analytical solution; confirmed <6% agreement across 3 superalloys
- Evaluated superalloys via parametric study, mesh independence testing; identified Rene 41 as optimal for shaft protection

CCGT Marine Propulsion System Design

February – April 2026

- Optimized variable-specific-heat Brayton cycle in MATLAB via parametric compression ratio and isentropic efficiency sweeps
- Maximized Rankine efficiency in Python via turbine pressure optimization with blade-erosion quality constraint
- Sized three-stage HRSG via Dittus-Boelter, Zukauskas, Rohsenow correlations, LMTD, and thermal resistance networks

SKILLS

- **Technical:** SolidWorks, Ansys, MATLAB, Python, Java, COMSOL, XFLR5, Arduino, Excel
- **Methods:** Computational Fluid Dynamics, Finite Element Analysis, AI-Assisted Workflows, GD&T
- **Equipment:** CNC, Lathe, Mill, Drill Press, Laser Cutter, 3D Printing, Band Saw, Soldering, Brazing

LEADERSHIP & MENTORING

Washington University in St. Louis | Material Science: *Teaching Assistant*

January – May 2026

- Lead weekly recitations and office hours for 160+ students, covering phase diagrams, mechanical properties, material selection
- Deliver constructive feedback on assignments and exams to reinforce student learning

Twain Elementary School | Math & Science: *Teaching Assistant*

August 2022 – May 2023

- Fostered student engagement by leading group discussions and individualized activities, earning recognition from supervisors
- Reduced staff preparation time through classroom setup and material organization projects

EXTRACURRICULARS

Pi Tau Sigma

March 2026 – Present

AIAA Design/Build/Fly & SAE Aero Design: *Lead Structures & Test Engineer, WashU Aircraft Team*

August 2023 – Present

Society of Hispanic Professional Engineers

August 2023 – Present